

MAHAD, Muhammad

✉ mmahad2-c@my.cityu.edu.hk ☎ (852) 5962 4263 in Muhammad Mahad 🌐 muhammadmahad4

EDUCATION

City University of Hong Kong - Full Tuition Scholarship (HKD\$140,000 p.a.) <i>BSc. in Computer Science</i> (CGPA: 3.93/4.3)	Sep 2023 – May 2027 Hong Kong
Alpha College - Full Tuition Scholarship <i>Cambridge A Levels</i> (3 A*s in Physics, Chemistry, Mathematics)	Aug 2021 – May 2023 Pakistan

WORK EXPERIENCE

Software Engineering Intern – Siemens Mobility Hong Kong	Jun 2025 – Present, Hong Kong
- Helped build a containerized, event-driven video analytics platform for intelligent transportation using Python (DeepStream), YOLOv8, and microservices (Spring Boot, Kafka, Redis, PostgreSQL), with deployments managed through Docker. - Developed an automated remote access solution using Hyper-V, CIS-hardened Ubuntu VMs (configured to CIS Benchmarks for enhanced security), and JumpServer (PAM platform deployed on Kubernetes), streamlining SSH key management and enforcing centralized, auditable firewall controls for secure team connectivity. - Implemented layered web security by automating IP threat mitigation with Fail2ban and granular request rate limiting via HAProxy stick tables, safeguarding production systems against DDoS and targeted attacks while maintaining robust logging and scalability. - Engineered and optimized multi-stage GitLab CI/CD pipelines for Python and Java projects on ARM and AMD platforms, achieving automated parallel builds, artifact management, and secure container deployments. - Built a full-stack image annotator tool with React, Java, and PostgreSQL to facilitate precise ROI drawing on camera streams, significantly improving data labeling efficiency for computer vision and data analytics projects.	
Team Lead, Technical Teaching Team (AWS Cloud Student Ambassador) – AWS Hong Kong	Dec 2025 – Present
- Lead the technical teaching team, designing and delivering 3 workshops on cloud and AI fundamentals, targeting the AWS AI Practitioner exam. - Coordinate with AWS professionals, faculty, and students to drive cloud awareness, organize training initiatives, and prepare students for AWS certifications.	

STEM Business Acceleration Intern – Youth Incubator Limited	Jun 2024 – Aug 2024, Hong Kong
- Scraped and analyzed 700+ URLs using targeted keywords, improving event data collection efficiency by 70%. Implemented a data pipelining workflow by applying prompt engineering with the OpenAI API for structured data extraction, analyzing results with Pandas, updating databases in MongoDB, and automating updates to the website using Python. - Developed a WhatsApp chatbot using the WhatsApp API, OpenAI API, and Botpress to automate inquiries, schedule meetings, and facilitate event registration by checking seat availability with the organizer, reducing manual workload by 20+ hours weekly. Deployed the chatbot server with Node.js and Express. - Engineered a secure login system, with JavaScript and React; Integrated Twilio and Memberstack for SMS/email verification, enhancing user security, and resulting in a 40% decrease in unauthorized access attempts.	

PERSONAL PROJECTS

Agentic AI for Cyber Security - <i>OpenClaw, LLMs, Docker, Kubernetes, React, Node.js</i>	<i>Final Year Project</i>
- Developing a secure, locally deployed agentic AI cybersecurity platform using OpenClaw and Hermes, Docker/Podman, and Kubernetes to automate workflows such as reconnaissance, log analysis, and controlled penetration testing in isolated lab environments. - Integrating LLMs and external tools (Nmap, AbuseIPDB) to execute automated threat correlation, while rigorously assessing the agent’s autonomy and mitigating inherent AI risks such as prompt injection and tool abuse.	
Auto-Emailing Website - <i>Full-stack Development / DeepSeek API, Google OAuth, React, Node.js, JavaScript, Express</i>	<i>GitHub</i> 
Built a Full-Stack email automation website in collaboration with a peer, leveraging the Google Cloud API for authentication and the DeepSeek API for AI-powered email extraction and content generation. Applied advanced prompt engineering to optimize AI responses, automating the mass emailing process by extracting emails, generating personalized messages, and sending them, streamlining job application workflows by 90%.	

EXTRACURRICULARS & AWARDS

Recipient of Belt and Road Scholarship (HKD\$140,000 p.a) – Hong Kong Government	Sep 2023 – May 2027
Placed 28th out of 150 teams in HKCERT Capture the Flag 2025 – HKCERT and DPO	Dec 2025
Participated in PolyU x NuttyShell Cybersecurity CTF 2026 – PolyU	Mar 2026
Tiger Award (Top 10 percent of Students) – City University	Sep 2023 – Present
Dean’s List (Sem A & B 23/24, Sem B 24/25, Sem A & B 25/26) – City University	Sep 2023 – June 2026
Teaching Assistant (CS2204: Fundamentals of Internet Applications Dev.) – City University	Aug 2025 – Present
Discipline Resident Tutor – Student Residence Office, CityU	Jan 2023
Habib University Meritorious Award – Habib University	Aug 2022 – May 2023
Executive Member – Alpha’s Robotics and Engineering Lab	Aug 2022 – May 2023
Head Planning Department – Alpha’s College STEM Society	Feb 2022 – July 2022
Winner of 3 National Science Olympiad Modules – Cedar College & Meritorious College	

LANGUAGES & SKILLS

Languages: Native Proficiency in Urdu and English, Spoken Proficiency in Hindi.
Technologies/Frameworks: Python, Java, JavaScript, TypeScript, C++, Bash, React, Node.js, Express, Spring Boot, CSS, MongoDB, PostgreSQL, Redis, AWS, Docker, Podman, Kubernetes, Apache Kafka, GitLab CI/CD, Git/GitHub, YOLOv8, Pandas, OpenAI/DeepSeek APIs, LLMs, HAProxy, Fail2ban, Nmap, JumpServer, Hyper-V, Linux, Jira, Visual Paradigm.
Soft Skills: Complex problem solving, Analytical thinking, Good interpersonal skills, Team player, Leadership skills, Innovative.
Hobbies: Table Tennis Player, Gaming (Led a Professional Sponsored team in PUBG Mobile), Hiking.